Framing Brief

Domain

AI applications and product practices in supply chain

AI is used more and more in supply chain work. Many vendors and consulting reports say AI can improve performance a lot. For example, they may claim higher forecast accuracy or lower stockouts. But these claims are not always supported by strong evidence. This project focuses on what AI is **really** used for in supply chains, and which practices have **credible evidence** of better performance.

Main Research Question

How is AI actually applied in supply chain operations, and which AI-driven practices have credible evidence of improving performance?

Sub-questions

1. **Which supply chain functions are most commonly supported by AI systems?**
   e.g., demand forecasting, inventory planning, routing, procurement, risk monitoring

2. **What types of AI techniques are used in these applications?**
   e.g., machine learning, deep learning, optimization, hybrid systems, LLM-based tools

3. **What performance improvements are claimed by vendors or practitioners?**
   e.g., higher accuracy, lower cost, fewer stockouts, faster decisions

4. **What empirical evidence supports or challenges these claims?**
   e.g., metrics, comparisons, experiments, before/after results, study design quality

5. **What limitations or risks are reported in real-world deployments?**
   e.g., data quality, model drift, explainability, integration cost, governance issues

Chosen tasks from the task menu

Task 1: Claim–Evidence Extraction

This task is useful because supply chain AI sources often contain strong claims. I will extract key claims and link each claim to direct text evidence. This helps measure whether the model can stay grounded and avoid making up support.

Task 2: Cross-source Synthesis

Different sources may disagree (for example, vendors may be optimistic while academic papers are cautious). This task helps compare sources, find agreements and disagreements, and highlight evidence gaps.

Models

- **Model A:** ChatGPT 5.2
- **Model B:** Claude Haiku 4.5

Test cases (Phase 1 plan)

I will use four test cases:

- **TC1 (Task 1):** An academic paper claiming AI-based demand forecasting improves forecast accuracy by X%.
- **TC2 (Task 1):** A consulting report or vendor white paper claiming AI inventory optimization reduces cost and/or stockouts.
- **TC3 (Task 2):** Paper A vs Paper B with different conclusions about the same AI method.
- **TC4 (Task 2):** Academic study vs company case study (academic is more cautious, company is more optimistic).

Expected outcome

At the end of Phase 1, I expect to have a clear framing that can be reused in later phases; prompts that encourage structured answers with evidence and citations and an evaluation sheet showing how the two models behave on the same tasks, including common failure modes.